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Origin and palaeomagnetism of the Mesoproterozoic Gangau tilloid (basal Vindhyan Supergroup), central India

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Abstract

The stratigraphic position and claimed glacial origin of the diamictitic 'Gangau tilloid' in central India have long been controversial. Our study shows that the ~ 50-m-thick tilloid overlies the Palaeoproterozoic Bijawar Group with angular unconformity and forms the base of the Mesoproterozoic (~ 1200–1400 Ma) Vindhyan Semri Group. The tilloid contains clasts up to 1 m across, virtually all of stable siliceous lithologies like those of the Bijawar Group, supported by a fine-grained haematitic matrix. Glacially faceted and striated clasts were not observed. The tilloid displays much disruption of beds, and pebble-sized clasts of lamellar and nodular chert–quartz underwent brittle fracture and plastic puckering during deposition. Values are high for SiO_2 (68.8–85.5%) and Fe_2O_3 (7.73–20.1%), and low for Al_2O_3 (1.03–5.50%) and other elements. The composition of the tilloid is similar to that of laterites formed on sandstone bedrock.

Thermal step demagnetisation of 91 core samples from seven sites spanning the tilloid revealed three components: a low-temperature Tertiary component A, an intermediate-temperature steep downward component B and, more rarely, a high-temperature less steep component C. Haematite is the likely carrier of the remanence, interpreted as chemical remanent magnetisation. C may be older than B and date from near the time of deposition. Bedding-corrected C has a direction of $D = 161.4°$, $I = 63.2°$ ($\alpha_{95} = 12.7°$) that gives a pole at 18.2°S, 93.4°E (dp = 15.8°, dm = 20.0°). The pole plots near 1200 Ma on the Australian Precambrian apparent polar wander path using the Veevers et al. (1991) reconstruction for East Gondwanaland.

The sedimentology, composition and geochemistry of the Gangau tilloid accord with deposition by continental debris-flows derived mainly from a ferruginous regolith formed on sedimentary rocks. Zoned chert–quartz bodies may have been precipitated in shrinkage cracks and voids in the lower, saturated zone of the regolith, and possibly also as thin beds or crusts of silica within the drainage system. The inferred ferruginous weathering in the source area favours warm and humid conditions with a dry season, and precipitation of silica in the regolith implies seasonal or longer wet and dry intervals. Debris-flow activity indicates brief episodes of abundant runoff. There is no evidence of glaciation. Production of abundant ferric iron in the source regolith implies the presence of appreciable atmospheric oxygen by ~ 1200–1400 Ma.

1. Introduction

The Proterozoic Vindhyan basin in central India (Fig. 1) extends over an area of nearly 200,000 km^2 (Mathur, 1982, 1989) and contains several diamictite units of alleged glacial origin. One such diamictite at the base of the Gangau Formation in Bundelkhand—informally termed here the 'Gangau tilloid', following Ahmad (1981)—has been persistently, although not unanimously, interpreted as glaciogenic. The stratigraphic placement of the Gangau tilloid also is controversial, different workers having assigned it

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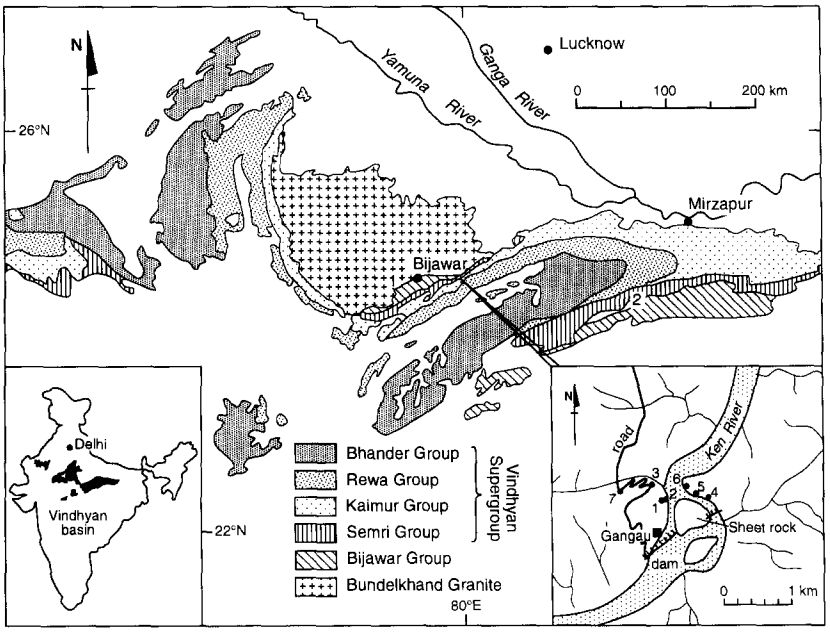


Fig. 1. Geological map of the Vindhyan basin, central India. *1* = Bundelkhand; *2* = Sidhi district, Son Valley. The inset at lower right shows the location of palaeomagnetic sites 1–7 in the Gangau tilloid near Gangau in Bundelkhand, Madhya Pradesh.

either to the top of the Palaeoproterozoic Bijawar Group or the base of the Mesoproterozoic Semri Group of the Vindhyan Supergroup.

In the Bijawar area of Bundelkhand, Vindhyan strata rest unconformably on a peneplained, weathered surface of the Archaean Bundelkhand Granite (Ghosh, 1981) or on the Bijawar Group. Here the Bijawar Group is tilted southward along the northern margin of the Vindhyan basin and comprises unmetamorphosed arenite, dolomite, limestone, jasper,

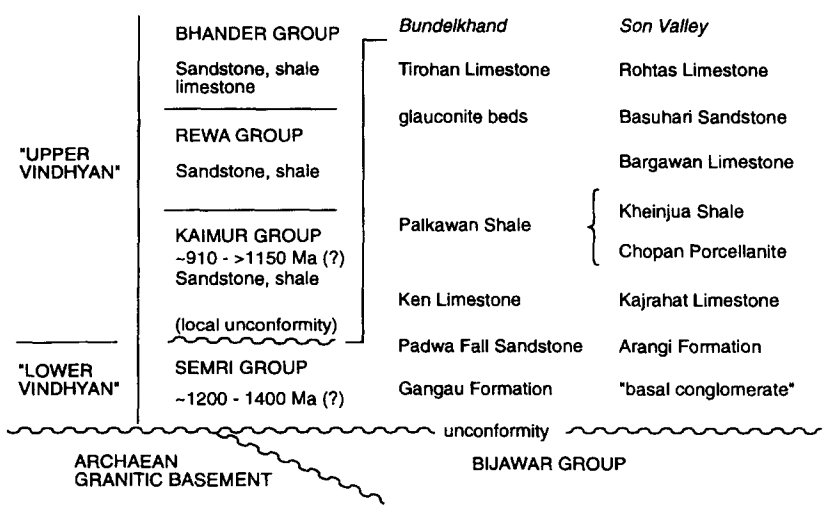


Fig. 2. Subdivision of the Vindhyan Supergroup, and stratigraphy and proposed correlations of the Semri Group in Bundelkhand and the Son Valley (adapted from Mathur, 1987). (Note: Padwa should read Pandwa.)

chert, basalt and breccia (Mathur, 1960; Mathur and Mani, 1978). Only the Bijawar Group contains in situ jasper in this area (Ahmad, 1958). In the Son Valley, the Bijawar and Semri groups dip steeply northward along the southern margin of the Vindhyan basin (Kailasam, 1976). Imprecise geochronological data suggest a Palaeoproterozoic age for the Bijawar Group. Crawford and Compston (1969) concluded from Rb–Sr dating that the age of the basalt must lie between 2400 and 2600 Ma. Mathur (1981a), however, compared the Bijawar Group with the Gwalior Group to the north, which has been dated at 1815 ± 200 Ma (Crawford and Compston, 1969).

The Vindhyan Supergroup is up to 5000–6000 m in thickness and comprises mainly sandstone, shale and limestone formations of shallow-water marine origin that are divisible into four groups (Fig. 2). The Semri Group, often referred to informally as the 'lower Vindhyan', is overlain with local angular unconformity by the 'upper Vindhyan' Kaimur, Rewa and Bhander groups. Most of the Vindhyan Supergroup is flat-lying to gently dipping and unmetamorphosed. The age range of the Vindhyan Supergroup is uncertain because of a paucity of material suitable for geochronology and the general lack of useful fossils (see Haldar and Ghosh, 1981; Mathur, 1982). K–Ar dating of glauconite from the Semri and Kaimur groups (Vinogradov et al., 1964) and K–Ar and Rb–Sr dating of kimberlites intrusive into those two groups (Crawford and Compston, 1969; Paul et al., 1975) suggest that the base of the Semri Group is at least ~ 1200 Ma and perhaps as much as 1400 Ma, and that the Kaimur Group is at least 910–940 Ma and possibly > 1150 Ma. The rarity of macrofossils in the Rewa and Bhander groups and the proximity of their mean pole position to that of Cambrian sediments of the Salt Range (McElhinny et al., 1978) together suggest a Neoproterozoic age for these topmost Vindhyan groups.

The Semri Group in Bundelkhand dips gently basinward along the northern margin of the Vindhyan basin. The Gangau Formation comprises diamictite, mudstone and ferruginous conglomerate and breccia (Mathur, 1981a), with the Gangau tilloid being exposed between arenites of the Bijawar and Semri groups along both banks of the Ken River near Gangau village. In this paper we investigate the stratigraphy, palaeoenvironment and palaeomagnetism of the Gangau tilloid, and also discuss the origin of other diamictites elsewhere in the Vindhyan basin that are claimed by some to be glaciogenic.

2. Claims of Palaeoproterozoic–Mesoproterozoic glaciation in India

A possible glacial origin for diamictites of the Bijawar and Semri groups has remained controversial since Oldham et al. (1901) compared a Precambrian boulder bed in the Son Valley with glacial boulder clays and late Palaeozoic tillites. This debate is briefly reviewed here.

2.1. Diamictite in the Bijawar Group, Son Valley

Oldham et al. (1901), in their pioneering study of Bijawar and Vindhyan strata in the Son Valley, drew attention to a conglomerate in the 'transitions' of the Sidhi district that comprises scattered rounded pebbles and boulders in a fine-grained argillaceous matrix, now slaty or schistose. They noted that the sorting of the rock was similar to that of glacial boulder clays of Europe and the Talchir (Gondwana) boulder bed in India, but added (p. 132) that "there are no other indications of glacial origin". This diamictite is part of the Parsoi phyllites, now believed to be equivalents of the Bijawar Group (Ahmad, 1971).

Ahmad (1955a) suggested that the deposit is a tillite because of the poor sorting and alleged presence of faceted and striated boulders. He observed, however, that striae are usually faint and well-striated clasts absent, and that faceted boulders are rare. In fact, the clasts illustrated by Ahmad (1955a) are tectonically deformed, and the single clast showing presumed striae appears to be sheared and stretched in the same direction as the surface markings, which suggest that the striae are of tectonic origin. Furthermore, Ahmad (1955a, p. 159) noted that the siliceous nature of the clasts "is not as varied as one would expect from a tillite" and that, given the large expanses of granitic basement, the absence of granitic clasts "is rather surprising". Overall, the evidence presented by Ahmad (1955a) for a glacial origin of this deposit is unconvincing and Ahmad (1981) himself seems subsequently to have abandoned the idea.

2.2. Basal diamictites of the Semri Group, Son Valley

Several early investigators suggested that certain poorly sorted beds in the 'basal conglomerate' of the Semri Group in the Son Valley are tillites (Dubey and Chaudhary, 1952; Law, 1954; Ahmad, 1955b; Dubey and Misra, 1956). The arguments for a glacial origin, namely the poor sorting, the angularity of clasts up to 25 cm in diameter and the presence of grains of fresh feldspar, are inconclusive and no unequivocal evidence for glaciation has been presented. Indeed, Auden (1933) and Law (1954) listed clasts of quartzite, quartz, jasper and chert (see also Mathur, 1981b), but not of granite, despite the basal conglomerate resting on granite basement. Granite clasts normally would be expected to be present if the beds are indeed tillites. The consensus among Indian geologists (e.g. Mathur, 1954, 1981b; Lakshmanan, 1968; Ahmad, 1971, 1981; Ghosh, 1981) is that these diamictites, which are laterally discontinuous and usually 3–8 m and rarely > 25 m in thickness, are part of the basal Vindhyan rudaceous Arangi Formation of non-glacial origin.

2.3. Gangau tilloid, Bundelkhand

The stratigraphic position and origin of the Gangau tilloid have long been debated by Indian geologists. Dubey and Chaudhary (1952) first drew attention to a "remarkable formation" consisting of angular and subangular clasts up to boulder size set in a fine-grained ferruginous matrix. They regarded the deposit as a typical tillite, noting that it occurred between the Bijawar and Semri groups in the same stratigraphic position as the "basal conglomerate" in the Son Valley. Chaudhary (1953) reiterated his view that the diamictite is a tillite, noting that the clasts show very distinct facets, rare striations, and a variety of lithologies including granite derived from the Bundelkhand Granite; however, he now placed the diamictite at the top of the Bijawar Group.

Mathur (1954) agreed that the diamictite at Gangau is a tillite and also placed it at the top of the Bijawar Group. He has since repeated these views (Mathur, 1960, 1981a, 1989), although in 1960 Mathur noted that he did not observe striations on the surface of the boulders. In the most detailed description of the Gangau Formation, Mathur (1981a) referred to the "Gangau Tillite" and stated (p. 429) that "Striations have been observed on the surface of a few boulders, most of which are also definitely faceted". He concluded that the diamictite appeared to be the only known Precambrian glaciogenic deposit in India.

Ahmad (1955b, 1958) initially regarded the diamictite of the Gangau Formation as glaciogenic and referred to it as the basal Semri "Gangau tillite". He observed (Ahmad, 1955b) that the presence of jasper pebbles in the Gangau Formation and of overlying passage beds are "hardly reconcilable with Bijawar age". Ahmad (1971) subsequently stated, however, that correlation of the "Gangau tillite" with the Bijawar or Vindhyan was "an open question". Later still, Ahmad (1981) concluded that no Proterozoic deposit in India has been established as glaciogenic, now referring to the "Gangau tilloid".

The Gangau tilloid therefore is the only Proterozoic diamictite within or adjacent to the Vindhyan basin for which a glacial origin has been persistently, albeit not unanimously, argued. The glaciogenic interpretation of the diamictite is based on its poor sorting together with the claimed occurrence of faceted and striated clasts. The latter two features are the only alleged characteristics of the deposit which, if present, would strongly support a glacial origin and their claimed occurrence is further discussed in Sect. 3.2.

3. The Gangau tilloid

The Gangau tilloid crops out along both banks of the Ken River (inset, Fig. 1) and at the northern end of 'Sheet rock', 300–700 m north of Gangau village and the nearby Gangau dam (24°38′N, 79°52′E). The best exposures are in a rock platform near low-stage river level on the eastern side of the river, in an adjacent prominent escarpment, and in a winding road cut 800 m north of Gangau. At all localities the tilloid is undeformed and shows no sign of metamorphism.

3.1. Contacts with Bijawar and Semri groups

The contact between the Bijawar Group and the Gangau Formation is exposed on the eastern bank of

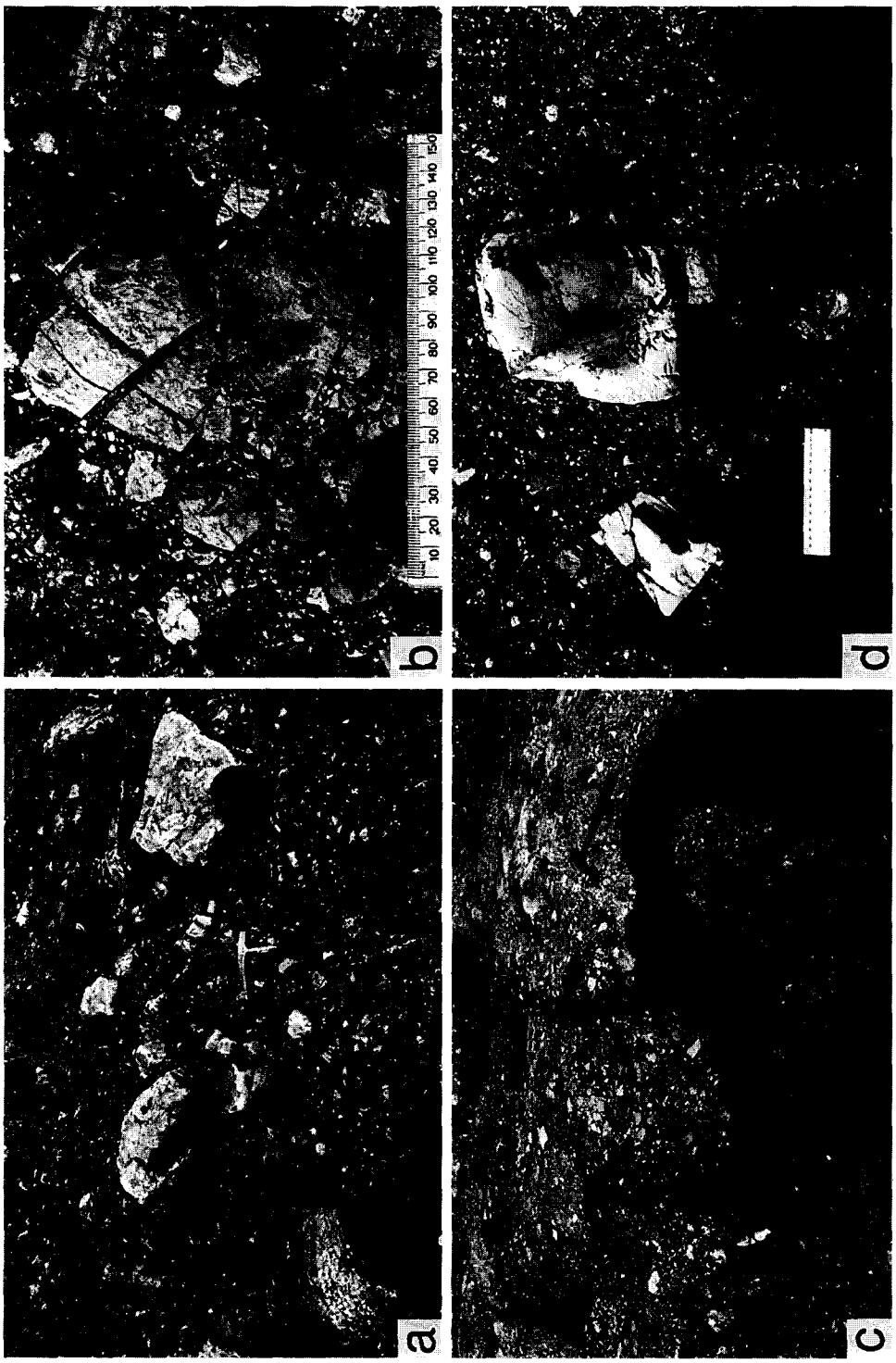


Fig. 3. Gangau tilloid. (a) Diamictite with large clasts of white quartzite (right of hammer), chert (left of hammer) and other siliceous rock fragments in a fine-grained haematitic matrix. Vertical section, hammer 33 cm long. (b) Fractured clasts of white quartzite showing minor separation of fragments. Vertical section, scale 15 cm. (c) Disturbed beds of haematitic sandstone and siltstone interbedded with diamictite, view looking down sequence over a near-horizontal surface. Haematitic siltstone evidently has been injected into overlying disrupted diamictite (foreground), and thin veins of haematitic siltstone cut the diamictite at centre left. Hammer 33 cm long. (d) Diamictite with two clasts of quartzite resting on a strongly eroded bed of haematitic siltstone; tongues of diamictite penetrate the siltstone at lower right. Near-horizontal section, scale 15 cm.

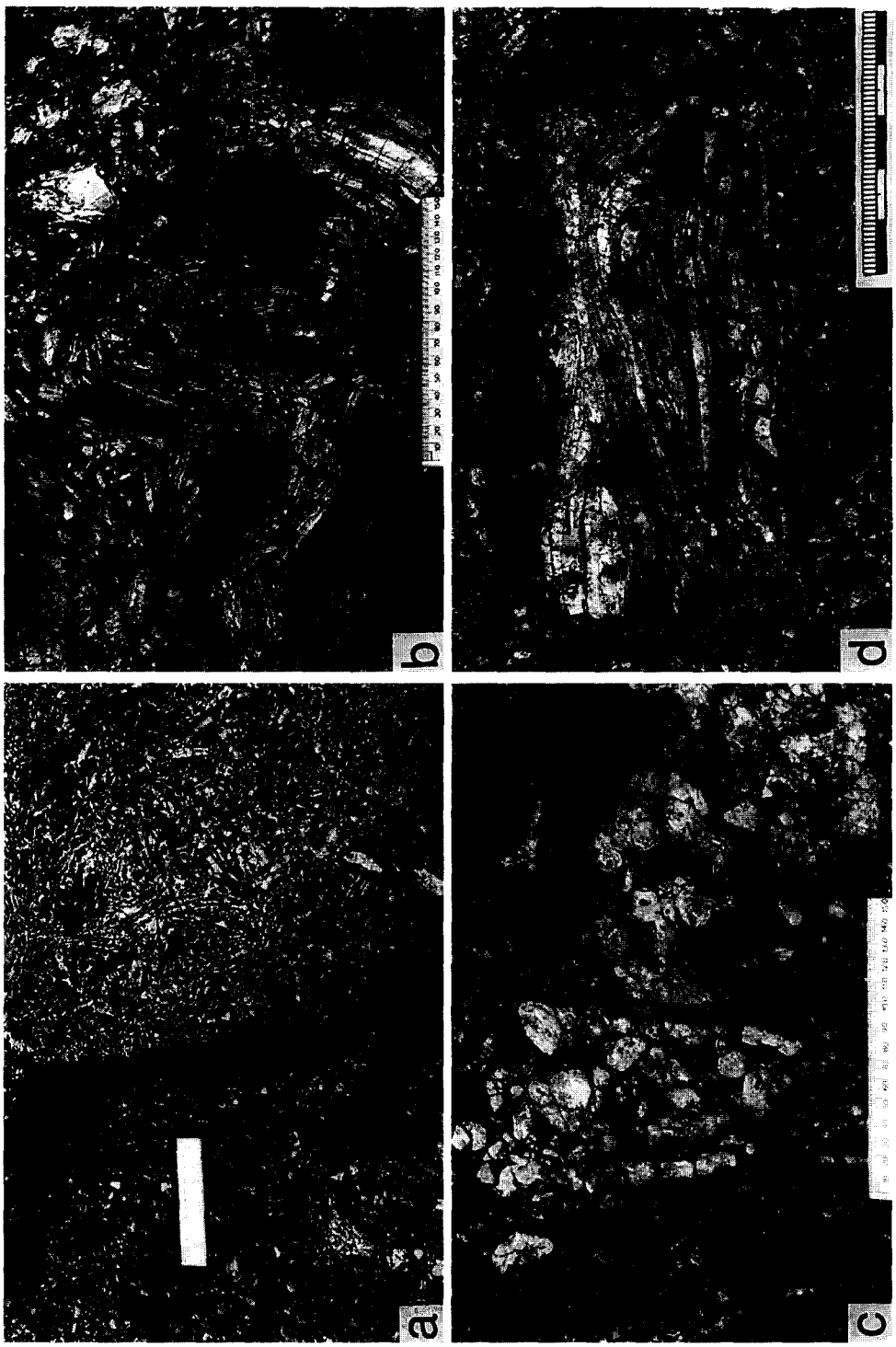


Fig. 4. Chert–quartz bodies in the Gangau tilloid. (a) Aggregate of lamellar chert–quartz fragments (right), some showing small-scale puckering, overlying crudely graded beds of fine-grained conglomerate, sandstone and haematitic siltstone (see also top left of Fig. 3c); scale 15 cm. (b) Lamellar chert–quartz showing evidence of brittle fracture, in haematitic siltstone and diamictite; scale 15 cm. (c) Nodular and lamellar chert–quartz showing some evidence of plastic deformation; scale 15 cm. (d) Aggregate of lamellar chert–quartz displaying plastic deformation and fracturing, and with internal spiral-like structure that suggests rolling during transport; scale 5 cm.

the Ken River 700 m north of the Gangau dam. There, feldspathic sandstone from the Amronia Quartzite of the Bijawar Group (Mathur and Mani, 1978) dips 36–37° southeastward and is overlain with angular unconformity by unstratified Gangau tilloid along a slightly irregular but near-horizontal contact. The usual red colour (5R 4–5/2) of the Amronia Quartzite at this locality is altered yellowish brown (10YR 6/4) to a depth of ~ 4 m below the unconformity, giving the rocks the appearance of a weathered horizon (colour designations from the Geological Society of America Rock-Color Chart, 1963). Fine-grained red (5R 3/2) sandstone of the Amronia Quartzite dipping 43° southeastward is exposed several metres below the base of the Gangau tilloid in the road cut north of Gangau, consistent with an angular unconformity.

The Pandwa Fall Sandstone, which overlies the Gangau tilloid with apparent conformity and dips 5–10° south-southeastward, is a medium- to fine-grained, well-sorted quartz arenite. Sedimentary structures include abundant trough and tabular cross-bed sets up to 1 m thick, local three-dimensional dune forms with superimposed ripples, ripple cross-bedding, flat-bedding with parting lineation, layers of polygonal and rounded mudstone intraclasts, and rare sole markings possibly after gypsum blades. The maturity and sedimentary structures of the Pandwa Fall Sandstone together typify sandstones of marginal-marine origin (e.g. Walker, 1984) and suggest an estuarine to subtidal environment, as proposed by Mathur (1989).

We conclude that the Gangau Formation forms the base of the Semri Group in Bundelkhand. The envisaged marginal-marine environment for the Pandwa Fall Sandstone and its apparent conformity with the Gangau tilloid imply that the latter accumulated near sea level.

3.2. Sedimentary features

The Gangau tilloid, which attains a thickness of about 50 m, consists mostly of diamictite (Figs. 3a, 3b) with clasts ranging from sand-grade to 1 m in diameter supported by a fine-grained, red (5R 4/2–3) matrix. Clasts are restricted to stable, siliceous rock types: fragments of white orthoquartzite are common, and other lithologies include coarse-grained metaquartzite, grey and brown chert, banded and non-banded jasper, quartz and, less commonly, medium- to coarse-grained red sandstone. Most of these rock types occur in the Bijawar Group. No clast of granite, gneiss, basalt or any other crystalline or igneous rock was observed by us. Mathur (1981a) noted that silicified conglomerate occurs near the top of the Gangau Formation, but our observations suggest that such rocks may be remnants of a Tertiary silicified duricrust. Silicified conglomerate is best developed on the western bank of the Ken River north of the Gangau dam and, as discussed in Sect. 4.2, palaeomagnetic analysis of diamictite samples from sites 1 and 2 in this area reveal a component ascribable to late Tertiary lateritisation.

Most clasts are angular to subangular; subrounded clasts are rare and only one rounded clast, a sandstone pebble, was observed. The clasts do not show any conspicuous or consistent preferred orientation. At several localities, pebble- and cobble-sized clasts were fractured in situ but show little separation or relative rotation of fragments (Fig. 3b). Such fracturing, which has similarity to the 'three-dimensional jigsaw puzzle effect' displayed by Holocene and modern landslide breccias (Shreve, 1966, 1968), may have resulted from collision between clasts. Many other clasts display incipient fractures and joints. These observations suggest that the angularity of clasts may partly reflect fragmentation during transport and deposition. Careful search was made for faceted and striated clasts indicative of glaciation, but without success.

The Gangau tilloid displays much evidence for the disruption, erosion and reworking of beds during deposition. Rare siltstone and sandstone interbeds < 1 cm to $\geqslant 30$ cm in thickness are strongly disturbed (Fig. 3c). The sandstone beds usually show gradational lower contacts with coarser-grained diamictite (Fig. 4a), but Bouma-type graded bedding was not observed. At one locality diamictite rests on a strongly eroded siltstone bed (Fig. 3d). In addition, the presence of subangular intraclasts of red mudstone and siltstone, and boulder-sized intraclasts of diamictite, indicate the erosion and reworking of partly indurated sediment. Rare round and elliptical bodies about 50 cm to 1 m in diameter, with cores of siltstone or diamictite and outer zones of sandy material, evidently are partly compacted intraclasts

that grew by accretion during transport. Such bodies have affinity with armoured mud balls rather than with till balls, which form by attrition and can be identified by their coarse grain-size and lack of internal structure (see Goldschmidt, 1994).

Distinctive clasts of white chert–quartz, consisting of lamellar forms usually $\leqslant 1$ cm in thickness and nodules $\leqslant 2$ cm in diameter, are scattered through the diamictite and locally form aggregates associated with crudely graded beds (Fig. 4a). These chert–quartz bodies show brittle fracture (Fig. 4b) and small-scale plastic deformation and puckering (Figs. 4a, 4c). The lamellar chert–quartz bodies typically show macroscopic transverse comb texture on opposite surfaces. In some clasts the comb texture is separated by a median band of haematitic material, suggesting expulsion of haematitic impurities during crystallisation of the quartz. The chert–quartz bodies locally occur with haematitic siltstone, and several lamellar bodies have the appearance of disrupted and fragmented beds (e.g. Figs. 4b, 4d). An aggregate of plastically deformed and fractured lamellar chert–quartz has an internal spiral-like structure (Fig. 4d), suggesting that some lamellar material was rolled into a compact body during transport.

3.3. Petrography

The matrix of the diamictite (Fig. 5a) comprises silt- to granule-grade fragments of quartz, chert, and other siliceous rocks set in fine-grained, opaque material containing much ultrafine haematitic pigment. Shapes of clastic grains are mostly angular to subangular and rarely rounded. Several rounded grains of quartz have optically-continuous overgrowths of secondary quartz that are rounded, indicating derivation from older sediments. Secondary veins of calcite and fine-grained micas occur locally. No fine-grained aggregate comparable to till pellets (Ovenshine, 1970) was observed. X-ray diffraction analysis confirmed that the matrix is dominantly quartz with substantial haematite, minor calcite and trace illite.

The chert–quartz bodies contain few impurities and typically are internally zoned. 'Megaquartz' with transverse comb texture commonly occurs on both sides of lamellar bodies and within nodules, and in some samples the chert–quartz is bordered by colloform chalcedonic 'microquartz' (Fig. 5b) (terminology of Knauth, 1994). Coarser-grained megaquartz with mosaic texture and showing outlines of euhedral faces occurs in the centre of some larger bodies.

Table 1
Whole-rock analyses of the Gangau tilloid

	1	2	3	4	5	6	7	8	9	10	11
Wt%:											
SiO_2	77.5	77.6	81.0	75.6	85.5	68.8	83.9	76.3	74.5	83.1	82.0
TiO_2	0.34	0.57	0.25	0.20	0.23	0.17	0.04	0.12	0.13	0.07	0.28
Al_2O_3	4.38	5.50	3.60	2.80	2.64	2.56	1.03	2.22	2.42	1.09	3.66
Fe_2O_3	12.2	10.9	9.53	13.4	7.73	18.1	7.83	13.2	20.1	12.6	7.92
FeO	0.20	0.20	0.20	0.20	0.20	< 0.10	0.20	0.20	0.20	0.20	0.52
MnO	0.03	0.04	0.05	0.13	0.04	0.29	0.03	0.11	0.14	0.04	0.02
MgO	0.15	0.30	0.16	0.20	0.12	0.24	0.06	0.18	0.17	0.09	0.94
CaO	0.05	0.07	0.56	2.78	0.04	3.74	1.83	2.34	0.55	0.44	0.12
Na_2O	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01
K_2O	1.17	1.42	0.93	0.57	0.65	0.42	0.21	0.32	0.30	0.21	0.97
P_2O_5	0.03	0.04	< 0.01	0.86	0.05	0.51	1.47	0.77	0.50	< 0.01	0.09
LOI	2.36	2.09	2.02	3.02	1.39	4.44	1.66	2.80	2.04	1.56	2.22
Ppm:											
Ce	31	40	26	19	20	36	7	17	24	21	–
Nd	10	18	13	5	12	13	6	4	7	6	–
La	14	17	9	5	8	10	1	3	3	6	–

All samples from outcrop by the Ken River north of Gangau dam. Samples: 1–5, diamictite matrix; 6–10, chert–quartz clasts in fine-grained matrix; 11, mudstone. Major elements determined by ICP (inductively-coupled plasma) and volumetric analysis at Amdel Laboratories Limited, Adelaide. Rare earth elements determined by X-ray fluorescence at the Department of Geology and Geophysics, University of Adelaide.

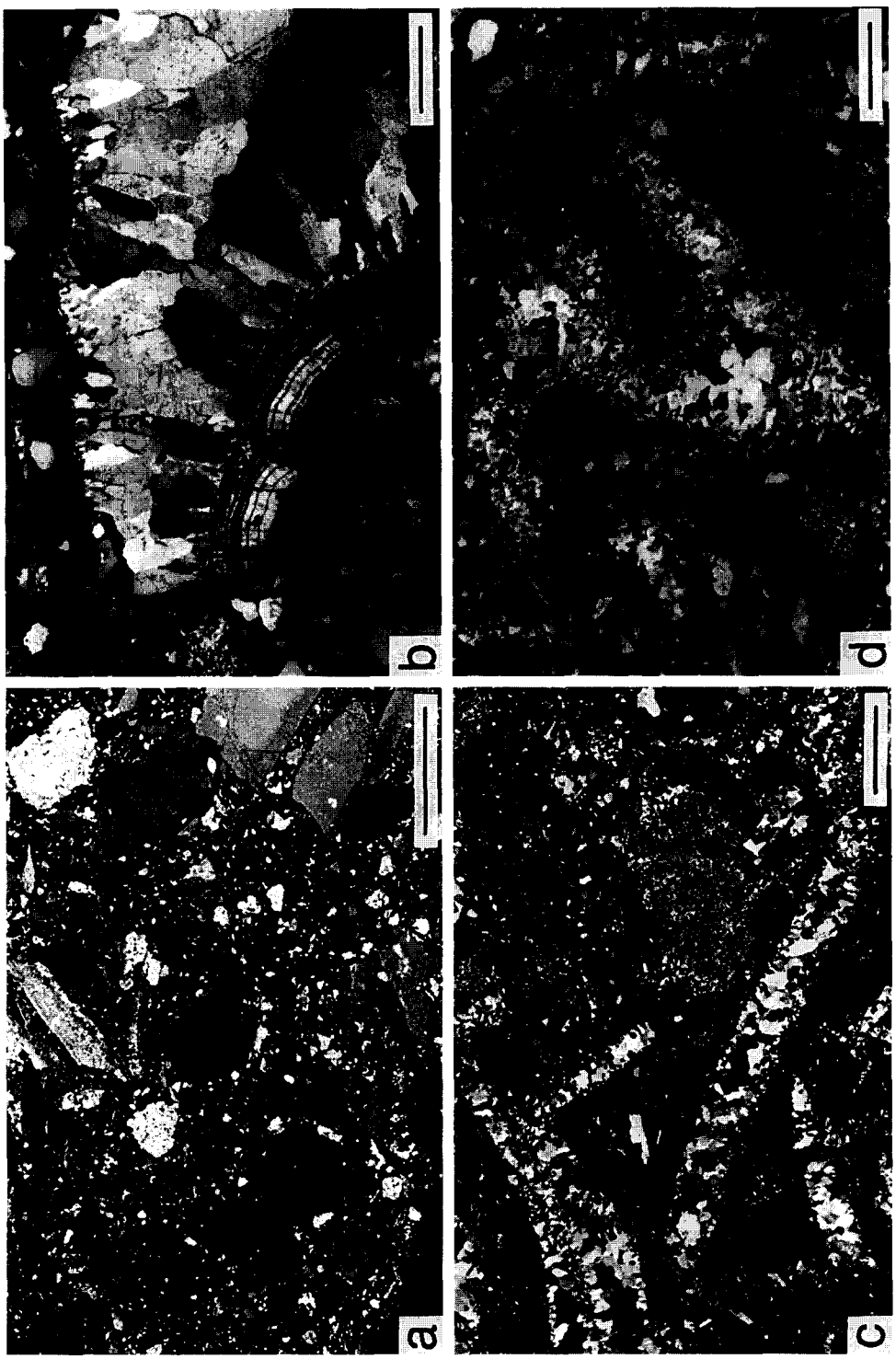


Fig. 5. Thin-sections of the Gangau tilloid, crossed polars. (a) Diamictite matrix, comprising mainly angular, silt- to granule-grade detrital fragments of quartz, chert, and other siliceous rocks set in fine-grained, opaque haematitic material; scale bar 4 mm. (b) Zoned lamellar chert–quartz showing colloform chalcedonic microquartz and megaquartz with comb and mosaic textures; scale bar 0.5 mm. (c) Lamellar chert–quartz fragments comprising megaquartz with delicate comb texture and mosaic texture, and showing evidence of brittle fracture and displacement of fragments; scale bar 4 mm. (d) Chert–quartz body of megaquartz with comb and mosaic textures, and showing evidence of plastic puckering; scale bar 0.5 mm.

One chert–quartz fragment was observed that comprised several lamellae of comb-textured megaquartz. The lamellar chert–quartz bodies commonly show evidence of brittle fracturing and displacement of fragments during deposition (Fig. 5c), and the curved shapes of some bodies indicate that plastic deformation and puckering also occurred at that time (Fig. 5d). The margins of the bodies usually are irregular, and show no evidence of abrasion and rounding.

The textures of the chert–quartz suggest that the silica was precipitated mainly as veins and cavity fill. The plastic deformation and puckering of some lamellar forms, evidently during deposition, implies the formation of at least some of the silica veins penecontemporaneously with accumulation of the Gangau tilloid, and the general lack of abrasion suggests local derivation.

3.4. Geochemistry

Whole-rock analyses of unweathered samples from the Gangau tilloid are given in Table 1. Samples comprised diamictite matrix with grains < 1 mm in diameter, chert–quartz fragments in a fine-grained haematitic matrix, and red mudstone. Combined SiO_2 and Fe_2O_3 values range from 86.9 to 95.7% for all samples. The very high SiO_2 values confirm the siliceous nature of clasts and matrix. Fe_2O_3 values are high, ranging from 7.73 to 20.1%, with generally higher values for samples containing chert–quartz clasts. Al_2O_3 values average only 2.90% for all samples, and samples with chert–quartz fragments consistently have the lower values. Values for FeO, MgO, Na_2O, K_2O and TiO_2 are low to negligible. Most CaO values are very low, with the moderate values of up to 3.74% in some samples evidently reflecting the presence of secondary calcite.

Mainly because of the low values for FeO, MgO and K_2O, the results in Table 1 are not comparable with compositions typical of iron-rich sedimentary rocks (James, 1966) or banded iron-formations (Trendall and Blockley, 1970). However, the low values for FeO, MgO, K_2O and Al_2O_3 and the high Fe_2O_3 content of the Gangau tilloid have similarities to values for some laterites (see Murthy et al., 1981; Schellmann, 1981). The high SiO_2 and low Al_2O_3 and TiO_2 values and the SiO_2/Al_2O_3 and $SiO_2/(Al_2O_3 + Fe_2O_3)$ ratios for the tilloid best agree with the composition of laterites developed on sandstone bedrock.

Values for the rare earth elements Ce, Nd and La are consistently low (Table 1). Positive Ce anomalies commonly are displayed by laterite profiles, particularly those developed on crystalline rocks, and negative Ce anomalies have been identified in laterite developed on sedimentary rocks and in coarse-grained marine sediments (see Banfield and Eggleton, 1989; Braun et al., 1990). The low values for these rare earth elements in the Gangau tilloid are consistent with a source terrain containing arenaceous sedimentary rocks.

$\delta^{18}O$ values for three chert–quartz clasts are +18.7‰, +22.6‰ and +23.2‰ (SMOW). The value of +18.7‰ is near the observed lower limit for Mesoproterozoic cherts (Knauth, 1992) and the other two values are near the middle of the Mesoproterozoic range. The low to moderate values may reflect the presence of meteoric water and/or relatively high climatic temperatures during precipitation of the chert–quartz and/or early diagenesis of the Gangau tilloid.

4. Palaeomagnetism of the Gangau tilloid

4.1. Methods and techniques

Routine palaeomagnetic methods (Collinson, 1983; Butler, 1992) were used throughout. Ninety-one oriented core samples were taken from outcrop at seven sites (inset, Fig. 1) spanning the stratigraphic thickness of the Gangau tilloid. Samples were oriented by magnetic compass and most samples also by sun compass. In the laboratory, at least one 21-mm-long specimen from each sample was subjected to thermal step demagnetisation and specimens from selected samples to alternating field (AF) step demagnetisation. Remanent magnetisations were measured using a CTF three-axis cryogenic magnetometer interfaced to a computer. The demagnetisers used were a Schonstedt AF demagnetiser model GSD-1 and the CSIRO automated three-stage carousel furnace, which is housed in a 4-m ten-coil Helmholtz set with automatic feedback maintaining a zero-field of < 5 nT.

Components of magnetisation were isolated using a WINDOWS' version of LINEFIND (Kent et al., 1983). In this procedure the linear segments are fitted to data points weighted according to the inverse of their measured variances. The variances are represented in plots of orthogonal projections by circles of different diameter.

4.2. Results

With few exceptions the directions of natural remanent magnetisation (NRM) of samples from the Gangau tilloid were spread from close to the dipole direction for the locality to a steep downward direction. The dipole declination is 0.0°, by definition,

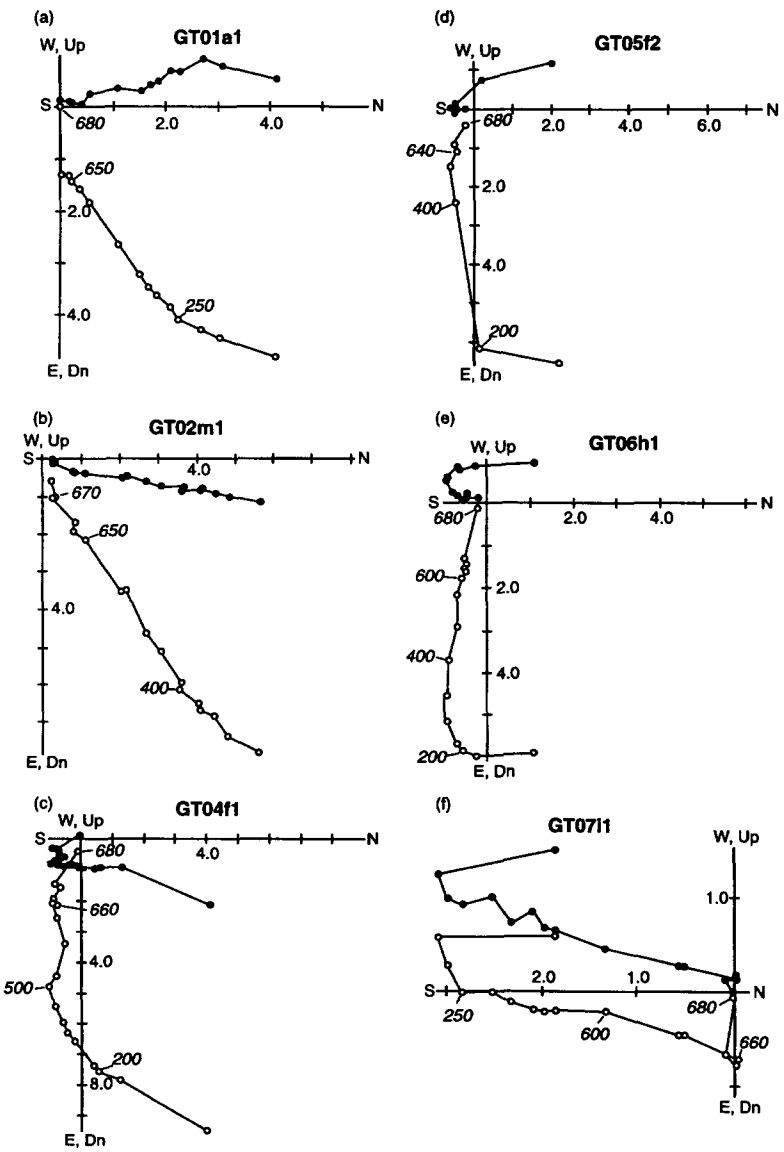


Fig. 6. Orthogonal projections showing examples of behaviour on thermal step demagnetisation of specimens of the Gangau tilloid. Steps include NRM, 150, 200, 250, 300, 350, 400, 450, 500, 550, 600, 640, 650, 660, 670, 675, 680 and 685°C (some steps omitted, or deleted after complete demagnetisation). Dots represent magnetisation vectors projected onto the horizontal plane and circles represent projections onto the vertical plane. Units are mA m^{-1} (= 1 μG = 10^{-6} emu/cc). The values plotted in italics next to data points refer to temperature (°C). *Dn* = down. (a) GT01a1 (site/sample/specimen). (b) GT02m1. (c) GT04f1. (d) GT05f2. (e) GT06hi. (f) GT07i1.

and dipole inclination is +42.5°. The present field, by comparison, has declination 0.5°W and inclination +36.2° using the 1985 International Geomagnetic Reference Field. The exceptions were almost the reverse of the dipole direction, suggesting some chemical remagnetisation during the late Tertiary (see Schmidt et al., 1983). Intensities of NRM were typically 5 to 20 mA m^{-1}.

Thermal step demagnetisation was found to be the most effective method of resolving the magnetic components present in the Gangau tilloid. Fig. 6 shows representative thermal demagnetisation plots. During such demagnetisation, directions in many specimens became progressively steeper until they were almost vertical downward; above 670°C the intensity became very weak and the remanence was completely blocked by 685°C. Specimen GT01a1 (Fig. 6a) typifies this behaviour. The thermal demagnetisation behaviour, coupled with the low NRM intensities, is consistent with the magnetic carrier being haematite. Magnetisation directions in many samples, especially from sites 1, 2 and 3, did not become much steeper than the NRM direction (Fig. 6b), suggesting that a relatively large proportion of late Tertiary magnetisation in these samples could not be preferentially demagnetised.

Samples from sites 4, 5 and 6 typically revealed three magnetisation components. After demagnetisation of the late Tertiary component A by 400° to 500°C, a steep downward magnetisation component B was revealed that gave way to a less steep southward component C at about 650°C (Figs. 6c–6e). Three components also were identified in some samples from site 7, but two of these components represent a dominantly recent viscous remanence and a reversed late Tertiary remanence; the remaining component was directed steeply downward (Fig. 6f).

Directions of components isolated by utilising LINEFIND are grouped in Fig. 7 for all samples, including components for fourteen samples of intraclasts and sandstone clasts (see below). Component A is well developed and readily identified at each site. At site 7 this component comprises both polarities, which may indicate a chemical remanent magnetisation from regional laterite development in the late Tertiary. Component B is fairly scattered, except at sites 4, 5 and 6, and is possibly contaminated by small amounts of component A. Component A was

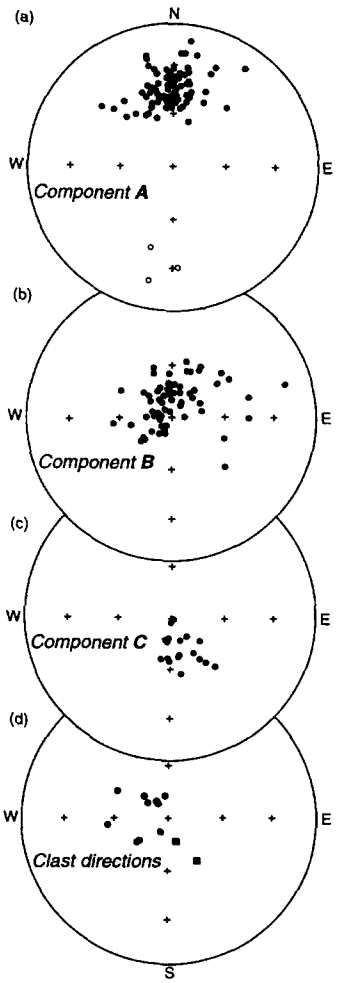


Fig. 7. Equal-area stereographic projections of in situ magnetisation component directions isolated for all samples from the Gangau tilloid by utilising LINEFIND. (a) Component A (500°C). (b) Component B (670°C). (c) Component C (675°C). (d) Directions for intraclasts and clasts only, component B plotted as circles and component C as squares. Dots refer to the lower hemisphere and circles to the upper hemisphere.

demagnetised by relatively low temperatures at sites 4, 5 and 6, which probably accounts for the clearer resolution of component B at these sites. The isolation of component C at sites 4, 5 and 6 suggests that the magnetic minerals present at these three sites are ideal palaeomagnetic recorders.

Eight core samples were taken from intraclasts of diamictite and mudstone in the Gangau tilloid at sites 4, 5 and 6, and a further six core samples from clasts

of red sandstone of uncertain provenance at sites 5 and 6. The intraclast samples displayed three components whose directions were not distinguishable from those of components A (seven directions), B (seven) and C (two). The sandstone samples displayed only two components, with directions similar to those of components A (six directions) and B (three). At face value the results suggest a negative conglomerate test (Fig. 7d), but since clasts and matrix alike would be subject to acquiring overprint components, such a test must be qualified as follows. If the C component is indeed the oldest, as interpreted here (see below), then the test is based on results from only two clasts. If the B component is an overprint then it is not surprising that it is carried by clasts. Nevertheless, the results are not inconsistent with a negative conglomerate test, whichever component is the oldest, which would rule out the presence of a detrital remanent magnetisation (DRM) acquired during deposition. The nature of the Gangau tilloid appears best in agreement with the 'Type A red bed' classification of Turner (1980), for which the mineralogy at the time of deposition is fairly mature and the stable remanence is an early chemical remanent magnetisation (CRM) acquired during diagenesis. The simplest approach would suggest that component C, with the higher blocking temperature, is older than component B and may date from close to the time of deposition.

Components B and C may require correction for the local mean tectonic dip of Semri Group bedding of 8° toward 160°, depending on the components' interpreted ages of acquisition. Table 2 lists the mean in situ directions of the components from each site, and the bedding-corrected directions for components B and C. Because the tilting of the Gangau Formation is almost certainly Proterozoic in age and the in situ component A is identical to the dipole field direction, bedding correction has been applied only to components B and C. The bedding-corrected component C has a mean declination of 161.4° and a mean inclination of 63.2° ($\alpha_{95} = 12.7°$) that give a pole at 18.2°S, 93.4°E (dp = 15.8°, dm = 20.0°). The results indicate a palaeolatitude of 44.7 ± 15.8° for the Gangau Formation, assuming that component C is a CRM acquired close to the time of deposition.

A comparison of the pole positions determined for components B and C (Table 2) with Australia's Precambrian apparent polar wander path (APWP) of Idnurm and Giddings (1988), using the East Gondwanaland reconstruction of Veevers et al. (1991), suggests that pole C may be a little younger than 1200 Ma and that pole B is somewhat younger than pole C, but older than 1100 Ma (Fig. 8). However, we cannot rule out older ages of ~ 1500 Ma for pole

Table 2
Gangau tilloid in situ and bedding-corrected palaeomagnetic directions and poles for the three components identified

Site	N	D_h (°)	I_h (°)	D_b (°)	I_b (°)	k	α_{95} (°)
Component A:							
1	16	3.0	49.4			80.6	4.1
2	16	6.8	46.8			43.8	5.6
3	11	4.0	30.9			35.8	7.7
4	9	355.0	41.4			52.4	7.2
5	18	348.6	47.6			48.0	5.0
6	13	347.6	43.6			14.8	11.2
7	10	358.3	37.2			21.4	10.7
Mean	7	357.7	42.6			90.3	6.4
Pole: lat. 87.7°N, long. 326.7°E (dp = 4.9°, dm = 7.9°)							
Component B:							
1	11	17.2	74.5	44.5	79.7	13.2	13.1
2	11	28.2	66.5	46.1	70.8	19.3	10.7
3	*4*	*84.0*	*50.5*	*88.3*	*46.8*	*6.89*	*37.8*
4	8	339.0	83.4	164.9	88.6	63.4	7.0
5	18	296.1	82.3	225.2	84.2	46.7	5.1
6	11	275.7	80.2	227.3	80.4	29.5	8.5
7	*3*	*69.8*	*83.8*	*122.0*	*79.9*	*21.0*	*27.6*
Mean	5	355.4	80.7	50.2	87.3	46.9	11.3
After bedding correction:							
Pole: lat. 27.4°N, long. 83.8°E (dp = 22.5°, dm = 22.5°)							
Rotated: lat. 33.7°S, long. 101.4°E (Veevers et al., 1991)							
Component C:							
4	6	153.4	63.7	154.8	55.8	60.5	8.7
5	7	179.1	71.4	173.7	63.7	99.8	6.1
6	6	154.2	77.2	156.4	69.2	37.2	11.1
Mean	3	161.8	71.2	161.4	63.2	95.9	12.7
After bedding correction:							
Pole: lat. 18.2°S, long. 93.4°E (dp = 15.8°, dm = 20.0°)							
Rotated: lat. 80.3°S, long. 98.7°E (Veevers et al., 1991)							

Notes: N, number of samples, or sites; D_h, I_h, mean declination and mean inclination with respect to present horizontal; D_b, I_b, mean declination and mean inclination with respect to bedding; k, precision parameter; α_{95}, semi-angle of cone of 95% confidence about mean direction; dp, dm, semi-major and semi-minor axes of polar error ellipse. Data shown in italics not used in calculation of mean values.

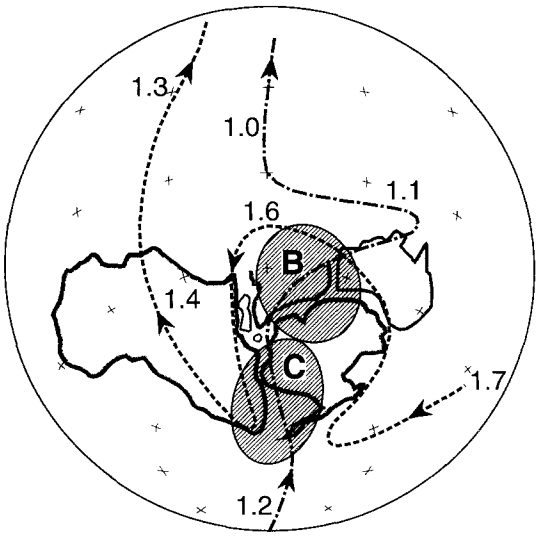


Fig. 8. Australian Mesoproterozoic APWP after Idnurm and Giddings (1988), with pole positions determined for components B and C of the Gangau tilloid (Table 2) transferred according to the finite rotations given by Veevers et al. (1991) for their East Gondwanaland reconstruction. Note that the path segments for 1.7–1.3 Ga (dashes, south pole path) and 1.2–1.0 Ga (dot–dashes, north pole path) have been drawn with opposite polarities to fit on the same hemisphere.

C and ~ 1600–1650 Ma for pole B. We prefer the younger ages of ~ 1200–1100 Ma, however, because these better accord with the estimated age of the base of the Semri Group and because they preserve the apparent relative ages of components C and B, assuming that the more stable C is the older.

The directions of components B and C differ from those determined for some Proterozoic kimberlitic rocks in India (Miller and Hargraves, 1994). However, the ages of the Gangau tilloid and the kimberlitic rocks and the timing of their respective magnetisations are not well constrained, and the diverse findings emphasise the need for more geochronological and palaeomagnetic data for Indian Proterozoic rocks.

5. Discussion

5.1. Chemical weathering in the source area

The matrix-supported texture of the Gangau tilloid and the presence of stable remanent magnetisation at some sites imply that the ferruginous matrix is an original constituent of the diamictite. The siliceous character of virtually all clasts, the presence of recycled quartz sand grains, and the geochemistry of the deposit together suggest derivation from a source terrain of sedimentary rocks, including sandstone, that experienced chemical weathering related to lateritisation. The Amronia Quartzite of the Bijawar Group forms the bedrock for the Gangau tilloid, and the data are consistent with derivation of the tilloid from a mature, weathered landsurface developed mainly on the Bijawar Group. Weathering in the source area is supported by observations that the Gangau tilloid overlies weathered Amronia Quartzite near the Ken River and that the Semri Group rests on weathered granite elsewhere in Bundelkhand (Ghosh, 1981). The siliceous character of rock fragments in the basal conglomerate of the Semri Group in the Son Valley (see Sect. 2.2) suggests that chemical weathering also occurred in the source area for the basal Vindhyan succession in that region.

The abundance in the Gangau tilloid of lamellar and nodular chert–quartz bodies, some of which may have formed penecontemporaneously with deposition of the tilloid, may provide further support for chemical weathering in the source area. The principal sources of dissolved silica in natural waters are volcanoes and thermal springs associated with volcanic activity, and the chemical weathering of silicate minerals. Andesite and pyroclastics do occur at the base of the Vindhyan Supergroup in Rajasthan 400 km west of Gangau (Prasad, 1984) and pyroclastics occur locally in the Semri Group in the Son Valley (Dubey and Misra, 1956; Ghosh, 1971), but no direct evidence of volcanic activity has been recognised in the Semri Group in Bundelkhand (Chaudhary, 1954) or in the Gangau tilloid. It has been suggested that porcellanites of the Palkawan Shale in Bundelkhand and the Chopan Porcellanite in the Son Valley (see Fig. 2) formed through the chemical alteration of highly siliceous tuffs (e.g. Oldham et al., 1901; Chaudhary, 1954; Law, 1954). However, Ghosh (1981) argued that the wide extent and uniformity of the porcellanites, the lack of any other significant products of volcanism, the general preponderance of silica cement in other Vindhyan rocks, and the presence of similar porcellanites in the Kaimur Group indicate rather that the continental land mass was the source of much of the silica.

In view of the independent evidence for chemical weathering in the source area of the Gangau tilloid, such weathering should be considered as a possible source of silica for the chert–quartz bodies. The solubility of amorphous silica in water is an order of magnitude greater than that of quartz and increases with temperature (Dove and Rimstidt, 1994), and weathering of rocks under warm climatic conditions would enhance the solution of amorphous silica. The $\delta^{18}O$ values for the chert–quartz clasts are consistent with warm conditions. The zoned lamellar chert–quartz with well developed comb texture (e.g. Fig. 5b) and nodular chert–quartz in the Gangau tilloid have strong textural similarities to silica or 'silcrete' lamellae and nodules in the lower parts of Tertiary ferruginous regolith profiles in the Paris Basin and the Central Massif in France (Thiry and Millot, 1987) and in central Australia (Thiry and Milnes, 1991). Silica gel or amorphous silica was precipitated in shrinkage cracks and voids at depth in the saturated zone of these profiles and subsequently recrystallised to chalcedony and microcrystalline and mosaic quartz. Thiry and Millot (1987) concluded that if some equilibrium exists between wet and dry seasonal or climatic periods, significant accumulation of silica occurs and the resultant silcrete is thick and extensive, as in the Eocene profiles of the Paris Basin.

Much of the chert–quartz in the Gangau tilloid therefore may have originated as silica gel and/or amorphous silica precipitated in shrinkage cracks and voids in the lower, saturated zone of the inferred ferruginous regolith mantling the source terrain. Crystallisation of chert and quartz evidently occurred both prior to and after incorporation of the silica bodies in the Gangau tilloid. Several lamellar chert–quartz bodies in the tilloid with the appearance of disrupted beds (Figs. 4b, 4d) may have been precipitated as thin beds or crusts of silica within the drainage system. Inorganic sedimentary precipitation of silica is rarely observed today and is confined to a few saline lakes (Peterson and von der Borch, 1965; Eugster, 1969). Siever (1957) noted, however, that because of a dearth of silica-secreting organisms, the concentration of silica in surface waters during much of the Precambrian may have been higher than seen today.

The inferred pre-Vindhyan ferruginous regolith may be preserved in places beneath the Semri Group in Bundelkhand and/or the Son Valley. Search for any such palaeosols is a worthy future project because of the possible implications concerning Mesoproterozoic atmospheric composition, but was beyond the scope of our study. The presence of oxidised palaeosols beneath Neoproterozoic (~ 750 Ma) alluvial fan deposits in northwest Scotland (Williams, 1968; Retallack and Mindszenty, 1994) suggests that oxygen was a substantial fraction of its present abundance by Neoproterozoic time (Retallack and Mindszenty, 1994). The inferred chemical weathering in the Gangau source area with the production of abundant ferric iron may extend such evidence for the presence of appreciable atmospheric oxygen back to ~ 1200–1400 Ma.

5.2. Deposition and palaeoenvironment of the Gangau tilloid

No evidence for a glacial origin of the Gangau tilloid was found during our study. Neither faceted nor striated clasts were observed by us, and argillite–dropstone facies is not present. Claims that the Gangau tilloid contains fragments of granite and faceted and striated clasts were made in the earliest descriptions (Dubey and Chaudhary, 1952; Chaudhary, 1953) and have been reiterated. Early workers may have mistaken planar fracture surfaces of clasts for faceted surfaces, and closely spaced fractures or joints in clasts for striations. Furthermore, the highly siliceous character of most clasts and the lack of clasts of granitoid and other crystalline or igneous rocks would not be expected if the Gangau tilloid were glaciogenic, because the Bundelkhand granitoid basement cropped out during Semri Group deposition and basalt occurs in the Bijawar Group. All clasts in the tilloid evidently were derived from the underlying Bijawar Group, with weathering of the source terrain explaining the lack of clasts of basalt. As noted above (Sections 2.1 and 2.2), similar reservations concerning glacial deposition may be raised for Proterozoic conglomerates in the Son Valley. Indeed, the thinness of basal coarse-grained rocks of the Semri Group, given their regional extent, better

accords with derivation from a widespread regolith than with glaciation.

The evidence provided by the Gangau tilloid for chemical weathering in the source area and vigorous deposition causing the disruption, erosion and reworking of beds, argues strongly for a continental debris-flow origin. 'Debris flow' is the term now generally applied to natural slurry flows that are marked by viscous frictional and inertial forces and continuous internal deformation (Takahashi, 1981; Clark, 1987; Pierson and Costa, 1987). Observed velocities of debris flows are from 0.5 to 20 m s^{-1}, and the momentum is transferred through particle collisions for the faster, inertial slurry flows. Conditions promoting continental debris-flows (Bull, 1972; Takahashi, 1981; Eisbacher, 1982; Church and Miles, 1987) include: (1) brief episodes of abundant water, usually intense rainfall but also abundant snowmelt; (2) unprotected slopes that are steeper than 15°; and (3) catchment areas underlain by deeply weathered bedrock and thick colluvial soils. A debris flow exerts enormous impact forces on obstacles in its path and the lateral pressures may crush even strong steel–concrete works. Consequently, a flow can cause severe erosion in the source area. Debris-flow deposits form cones, fans, sheets and lobate tongues in or beyond canyons on surfaces whose gradients are $\leqslant 3°$. Typically, the deposits are poorly sorted, matrix-supported diamictites, comprising a fine-grained muddy matrix and angular to subangular rock fragments that range in size from sand and fine-grained gravel up to boulders weighing many tonnes (Bull, 1972; Kostaschuk et al., 1986; Jackson et al., 1987). Bedding is normally poorly defined, although graded bedding may result from more fluid flows and bedding planes may occur between flows.

Deposition by continental debris-flows derived from an upland catchment area of weathered rocks can explain the important sedimentary features of the Gangau tilloid: (1) the poor sorting, with clasts up to 1 m in diameter supported by a fine-grained haematitic matrix; (2) the angular to subangular shape and stable lithologies of clasts; and (3) the abundant evidence for vigorous deposition such as disturbed and disrupted bedding, local erosional contacts between beds, in situ fracturing of clasts evidently by particle collisions, and the occurrence of intraclasts. The presence of mud-ball-like bodies, lamellar chert–quartz that evidently was fractured and deformed during deposition, and local graded beds are consistent with vigorous transport and rapid deposition.

The Gangau tilloid provides evidence of a variable climate in the source area. According to Tardy and Roquin (1992), ferricretes best develop under seasonally contrasted tropical climates with a mean annual temperature of $\sim 30°C$ and 1300–1700 mm annual rainfall. They concluded that the formation of haematite in soil profiles is favoured by increase in temperature, a dry season, and upslope well-drained conditions. The inferred occurrence of a ferruginous regolith in the Gangau source area therefore suggests warm and humid conditions with seasonal dryness. The presence of detrital chert–quartz bodies in the Gangau tilloid, many of which may have been derived from silcrete veins and nodules in the lower part of the source regolith, also suggests seasonal or longer wet and dry intervals and a fluctuating water table in the source area (see Thiry and Millot, 1987). Some silica may have been precipitated as thin beds or crusts within the drainage system, possibly through evaporation or bacterial action (see Peterson and von der Borch, 1965; Knauth, 1994). Brief episodes of abundant water runoff triggered vigorous debris-flow activity that eroded the regolith material and any surficial crusts of silica.

Without vegetative cover to stabilise the regolith profile, a stable landsurface of low topographic relief seems required for such weathering in the source area. Erosion of this weathered landsurface and debris-flow activity may have been initiated by earth movements at the margin of the developing Vindhyan basin. These events evidently occurred near sea level in middle palaeolatitudes, assuming that component C was acquired close to the time of deposition.

The Gangau tilloid contrasts with ferruginous diamictites of Neoproterozoic age such as the haematitic Sayunei Formation in northwestern Canada and the locally iron-rich Chuos Formation in Namibia (Hambrey and Harland, 1981; Henry et al., 1986; Bühn et al., 1992). These Neoproterozoic diamictites are basin-wide in extent and up to several hundred metres thick, display faceted and striated clasts and numerous lonestones, and contain extrabasinal clasts of crystalline basement rocks. Such

deposits are best regarded as glaciomarine, in places reworked by submarine mass flows, with a possible exhalative origin for the iron in ferruginous beds.

6. Conclusions

The main findings of our study are:

(1) The ~ 50-m-thick diamictitic 'Gangau tilloid' in Bundelkhand, central India, overlies sandstones of the Palaeoproterozoic Bijawar Group with angular unconformity, and is the lowermost unit of the Mesoproterozoic (~ 1200–1400 Ma) Semri Group at the base of the Vindhyan Supergroup.

(2) The main features of the Gangau tilloid, namely poor sorting, angular to subangular shapes and stable lithologies of most clasts, evidence for vigorous deposition and disruption of bedding, haematitic matrix, and presence of silcrete-like lamellar and nodular chert–quartz bodies, accord with deposition by continental debris-flows derived mainly from a ferruginous and locally silicified regolith formed on sedimentary rocks. The tilloid displays no evidence of glaciation.

(3) The inferred occurrence of ferruginous regolith and silcrete in the source area suggests warm and humid conditions with seasonal dryness, a stable landsurface of low relief, and the presence of appreciable atmospheric oxygen. Debris-flow activity implies brief episodes of abundant water-runoff.

(4) Haematite is the likely carrier of the remanent magnetisation, which is interpreted as CRM. The most stable component of magnetisation, component C, has a bedding-corrected mean declination of 161.4° and a mean inclination of 63.2° that give a pole at 18.2°S, 93.4°E. This pole plots near 1200 Ma on the Australian Precambrian APWP of Idnurm and Giddings (1988) using the Veevers et al. (1991) reconstruction for East Gondwanaland.

(5) The Gangau tilloid accumulated near sea level at the margin of the Vindhyan basin in middle palaeolatitudes, assuming that component C was acquired close to the time of deposition.

Our findings do not support the conclusion (e.g. Hambrey and Harland, 1981, p. 943) that the Gangau tilloid records Palaeoproterozoic glaciation in India. Indeed, there is no confirmed occurrence of Palaeoproterozoic or Mesoproterozoic glaciation anywhere on the Subcontinent.

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References

Ahmad, F., 1955a. An ancient tillite in central India. Q. J. Geol. Mining Metall. Soc. India, 27: 157–161.

Ahmad, F., 1955b. Glaciation in the Vindhyan System. Current Sci., 24: 231.

Ahmad, F., 1958. Palaeogeography of central India in the Vindhyan Period. Rec. Geol. Surv. India, 87: 513–548.

Ahmad, F., 1971. Geology of the Vindhyan System in the eastern part of the Son Valley in Mirzapur District, U.P. Rec. Geol. Surv. India, 96: 1–41.

Ahmad, F., 1981. Some problems in the palaeogeography of the Vindhyan times in central India. In: Proc. Symp. Vindhyans of Central India. Geol. Surv. India Misc. Publ., 50: 69–74.

Auden, J.B., 1933. Vindhyan sedimentation in the Son Valley, Mirzapur District. Geol. Surv. India Mem., 62(2): 141–250.

Banfield, J.F. and Eggleton, R.A., 1989. Apatite replacement and rare earth mobilization, fractionation, and fixation during weathering. Clays Clay Miner., 37: 113–127.

Braun, J.-J., Pagel, M., Muller, J.P., Bilong, P., Michard, A. and Guillet, B., 1990. Cerium anomalies in lateritic profiles. Geochim. Cosmochim. Acta, 54: 781–795.

Bühn, B., Stanistreet, I.G. and Okrusch, M., 1992. Late Proterozoic outer shelf manganese and iron deposits at Otjosondu (Namibia) related to the Damara oceanic opening. Econ. Geol., 87: 1393–1411.

Bull, W.B., 1972. Recognition of alluvial-fan deposits in the stratigraphic record. In: J.K. Rigby and W.K. Hamblin (Editors), Recognition of Ancient Sedimentary Environments. Soc. Econ. Paleontol. Mineral. Spec. Publ., 16: 63–83.

Butler, R.F., 1992. Paleomagnetism: Magnetic Domains to Geologic Terranes. Blackwell, Oxford, 319 pp.

Chaudhary, M.S., 1953. Late Pre-Cambrian glaciation and its relationship with the Bijawar series in the type area of Bundelkhand. Proc. 40th Indian Sci. Congr., 3: 20.

Chaudhary, M.S., 1954. On the occurrence of Porcellanite in the Semri Series (Type area). Q. J. Geol. Mining Metall. Soc. India, 26: 49–53.

Church, M. and Miles, M.J., 1987. Meteorological antecedents to debris flow in southwestern British Columbia; some case histories. In: J.E. Costa and G.F. Wieczorek (Editors), Debris Flows/Avalanches: Process, Recognition, and Mitigation. Geol. Soc. Am. Rev. Eng. Geol., 7: 63–79.

Clark, G.M., 1987. Debris slide and debris flow historical events in the Appalachians south of the glacial border. In: J.E. Costa and G.F. Wieczorek (Editors), Debris Flows/Avalanches: Process, Recognition, and Mitigation. Geol. Soc. Am. Rev. Eng. Geol., 7: 125–138.

Collinson, D.W., 1983. Methods in Rock Magnetism and Palaeomagnetism. Chapman and Hall, London, 503 pp.

Crawford, A.R. and Compston, W., 1969. The age of the Vindhyan System of Peninsular India. Q. J. Geol. Soc. London, 125: 351–371.

Dove, P.M. and Rimstidt, J.D., 1994. Silica–water interactions. In: P.J. Heaney, C.T. Prewitt and C.V. Gibbs (Editors), Silica. Mineralogical Society of America, Washington, D.C., pp. 259–308.

Dubey, V.S. and Chaudhary, M.S., 1952. Late Pre-Cambrian glaciation in central India. Current Sci., 21: 331–332.

Dubey, V.S. and Misra, K.K., 1956. Geology of the eastern part of the Son Valley, with special reference to the acid-igneous activity of the Lower Vindhyans. Proc. 43rd Indian Sci. Congr., 3: 178–179.

Eisbacher, G.H., 1982. Mountain torrents and debris flows. Episodes, Vol. 1982(4): 12–17.

Eugster, H.P., 1969. Inorganic bedded cherts from the Magadi area, Kenya. Contr. Mineral. Petrol., 22: 1–31.

Ghosh, D.B., 1981. The Vindhyan Basin in Bundelkhand–Son Valley region. In: Proc. Symp. Vindhyans of Central India. Geol. Surv. India Misc. Publ., 50: 75–88.

Ghosh, S.M., 1971. Petrology of the Porcellanite rocks of the Samaria area, Sidhi District, Madhya Pradesh. Q. J. Geol. Mining Metall. Soc. India, 43: 153–164.

Goldschmidt, P.M., 1994. Armoured and unarmoured till balls from the Greenland Sea Floor. Mar. Geol., 121: 121–128.

Haldar, D. and Ghosh, D.B., 1981. A discussion on the uncertainties in the dating of the Vindhyan and Bijawar rocks. In: Proc. Symp. Vindhyans of Central India. Geol. Surv. India Misc. Publ., 50: 223–228.

Hambrey, M.J. and Harland, W.B. (Editors), 1981. Earth's pre-Pleistocene Glacial Record. Cambridge University Press, Cambridge, 1004 pp.

Henry, G., Stanistreet, I.G. and Maiden, K.J., 1986. Preliminary results of a sedimentological study of the Chuos Formation in the central zone of the Damara Orogen: evidence of mass flow processes and glacial activity. Commun. Geol. Surv. S.W. Africa/Namibia, 2: 75–92.

Idnurm, M. and Giddings, J.W., 1988. Australian Precambrian polar wander: a review. Precambrian Res., 40/41: 61–88.

Jackson, L.E., Kostaschuk, R.A. and MacDonald, G.M., 1987. Identification of debris flow hazard on alluvial fans in the Canadian Rocky Mountains. In: J.E. Costa and G.F. Wieczorek (Editors), Debris Flows/Avalanches: Process, Recognition, and Mitigation. Geol. Soc. Am. Rev. Eng. Geol., 7: 115–124.

James, H.L., 1966. Chemistry of the iron-rich sedimentary rocks. U.S. Geol. Surv. Prof. Pap., 440–W: 1–61.

Kailasam, L.N., 1976. Geophysical studies of the major sedimentary basins of the Indian craton, their deep structural features and evolution. Tectonophysics, 36: 225–245.

Kent, J.T., Briden, J.C. and Mardia, K.V., 1983. Linear and planar structure in ordered multivariate data as applied to progressive demagnetization of palaeomagnetic remanence. Geophys. J. R. Astron. Soc., 75: 593–621.

Knauth, L.P., 1992. Origin and diagenesis of cherts: an isotopic perspective. In: N. Clauer and S. Chaudhuri (Editors), Isotopic Signatures and Sedimentary Rocks. Springer-Verlag, Berlin, pp. 123–152.

Knauth, L.P., 1994. Petrogenesis of chert. In: P.J. Heaney, C.T. Prewitt and C.V. Gibbs (Editors), Silica. Mineralogical Society of America, Washington, D.C., pp. 233–258.

Kostaschuk, R.A., MacDonald, G.M. and Putnam, P.E., 1986. Depositional process and alluvial fan–drainage basin morphometric relationships near Banff, Alberta, Canada. Earth Surface Processes Landforms, 11: 471–484.

Lakshmanan, S., 1968. On the nature of the basal conglomerate of the Semri Series in the Son Valley. Proc. Ntl. Inst. Sci. India, A, 34: 50–55.

Law, Y.D., 1954. Contributions to the geology of Son Valley in Vindhya Pradesh. Q. J. Geol. Mining Metall. Soc. India, 26: 65–79.

Mathur, S.M., 1954. Late Pre-Cambrian glaciation in central India —a rejoinder. Current Sci., 23: 7–8.

Mathur, S.M., 1960. A note on the Bijawar Series in the eastern part of the type area. Rec. Geol. Surv. India, 86: 539–544.

Mathur, S.M., 1981a. The Middle Proterozoic Gangau Tillite, Bijawar Group, Central India. In: M.J. Hambrey and W.B. Harland (Editors), Earth's pre-Pleistocene Glacial Record. Cambridge University Press, Cambridge, pp. 428–430.

Mathur, S.M., 1981b. Basal Vindhyan diamictite in the Son Valley, Central India. In: M.J. Hambrey and W.B. Harland (Editors), Earth's pre-Pleistocene Glacial Record. Cambridge University Press, Cambridge, pp. 424–427.

Mathur, S.M., 1982. Precambrian sedimentary sequences of India: their geochronology and correlation. Precambrian Res., 18: 139–144.

Mathur, S.M., 1987. Geochronology and biostratigraphy of the Vindhyan Supergroup. Geol. Surv. India Spec. Publ., No. 11, Vol. 1: 23–44.

Mathur, S.M., 1989. Sedimentary environmental analysis of the Precambrian basins in Peninsular India. Proc. 5th Indian Assoc. Sedimentol., 1985. J. Indian Assoc. Sedimentol., 8: 103–120.

Mathur, S.M. and Mani, G., 1978. Geology of the Bijawar Group in the type area, Madhya Pradesh. Proceedings of the Symposium on The Purana Formations of Peninsular India, University of Saugar, Sagar, pp. 313–320.

McElhinny, M.W., Cowley, J.A. and Edwards, D.J., 1978. Palaeomagnetism of some rocks from Peninsular India and Kashmir. Tectonophysics, 50: 41–54.

Miller, K.C. and Hargraves, R.B., 1994. Paleomagnetism of some Indian kimberlites and lamproites. Precambrian Res., 69: 259–267.

Murthy, M.K., Kalsotra, M.R. and Sheo Prasad, 1981. Bauxite over the Vindhyan formations of India—a study. In: Lateritisation Processes. Proc. Int. Seminar on Lateritisation Processes, Trivandrum, India, December 1979. Oxford and Ibh Publishing Co., New Delhi, pp. 193–201.

Oldham, R.D., Datta, P.N. and Vredenburg, E. 1901. Geology of

the Son Valley in the Rewah State and of parts of the adjoining districts of Jabalpur and Mirzapur. Geol. Surv. India Mem., 3: 1–178.

Ovenshine, A.T., 1970. Observations of iceberg rafting in Glacier Bay, Alaska, and the identification of ancient ice-rafted deposits. Geol. Soc. Am. Bull., 81: 891–894.

Paul, D.K., Rex, D.C. and Harris, P.G., 1975. Chemical characteristics and K–Ar ages of Indian kimberlite. Geol. Soc. Am. Bull., 86: 364–366.

Peterson, M.N.A. and von der Borch, C.C., 1965. Chert: modern inorganic deposition in a carbonate-precipitating locality. Science, 149: 1501–1503.

Pierson, T.C. and Costa, J.E., 1987. A rheologic classification of subaerial sediment–water flows. In: J.E. Costa and G.F. Wieczorek (Editors), Debris Flows/Avalanches: Process, Recognition, and Mitigation. Geol. Soc. Am. Rev. Eng. Geol., 7: 1–12.

Prasad, B., 1984. Geology, sedimentation and palaeogeography of the Vindhyan Supergroup, southeastern Rajasthan. Geol. Surv. India Mem., 116: 1–148.

Retallack, G.J. and Mindszenty, A., 1994. Well preserved late Precambrian paleosols from northwest Scotland. J. Sediment. Res., A64: 264–281.

Schellmann, W., 1981. Considerations on the definition and classification of laterites. In: Lateritisation Processes. Proc. Int. Seminar on Lateritisation Processes, Trivandrum, India, December 1979. Oxford and Ibh Publishing Co., New Delhi, pp. 1–10.

Schmidt, P.W., Prasad, V. and Ramam, P.K., 1983. Magnetic ages of some Indian laterites. Palaeogeogr., Palaeoclimatol., Palaeoecol., 44: 185–202.

Shreve, R.L., 1966. Sherman landslide, Alaska. Science, 154: 1639–1643.

Shreve, R.L., 1968. The Blackhawk landslide. Geol. Soc. Am. Spec. Pap., 108, 47 pp.

Siever, R., 1957. The silica budget in the sedimentary cycle. Am. Mineral., 42: 821–841.

Takahashi, T., 1981. Debris flow. Ann. Rev. Fluid Mech., 13: 57–77.

Tardy, Y. and Roquin, C., 1992. Geochemistry and evolution of lateritic landscapes. In: I.P. Martini and W. Chesworth (Editors), Weathering, Soils and Paleosols. Developments in Earth Surface Processes 2, Elsevier, Amsterdam, pp. 407–443.

Thiry, M. and Millot, G., 1987. Mineralogical forms of silica and their sequence of formation in silcretes. J. Sediment. Petrol., 57: 343–352.

Thiry, M. and Milnes, A.R., 1991. Pedogenic and groundwater silcretes at Stuart Creek opal field, South Australia. J. Sediment. Petrol., 61: 111–127.

Trendall, A.F. and Blockley, J.G., 1970. The iron formations of the Precambrian Hamersley Group, Western Australia. Geol. Surv. W. Aust. Bull., 119, 366 pp.

Turner, P., 1980. Continental Red Beds. Developments in Sedimentology 29, Elsevier, Amsterdam, 562 pp.

Veevers, J.J., Powell, C.McA. and Roots, S.R., 1991. Review of seafloor spreading around Australia, I. Synthesis of the patterns of spreading. Aust. J. Earth Sci., 38: 373–389.

Vinogradov, A., Tugarinov, A., Zhykov, C., Stapnikova, N., Bibikova, E. and Khorre, K., 1964. Geochronology of Indian Pre-Cambrian. Rep. 22nd Int. Geol. Congr., New Delhi, 10: 553–567.

Walker, R.G., 1984. Shelf and shallow marine sands. In: R.G. Walker (Editor), Facies Models (2nd ed.). Geological Association of Canada, Ontario, pp. 141–170.

Williams, G.E., 1968. Torridonian weathering, and its bearing on Torridonian palaeoclimate and source. Scott. J. Geol., 4: 164–184.